

On-orbit Performance Evaluation of AQUARIUS: a Water Resistojet Propulsion System during Initial Flight Operation of a 6U CubeSat EQUULEUS*

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A 6U CubeSat EQUULEUS (EQUilibrium Lunar-Earth point 6U Spacecraft) successfully executed an orbit transfer using a water-propelled propulsion system in deep space for the first time using AQUARIUS (AQUA Resistojet propUlsion System), a water resistojet propulsion system. We report on-orbit performance analysis of AQUARIUS in its initial flight operation. AQUARIUS employed a distinct water droplet vaporization structure for efficient gas-liquid separation and a power-saving strategy for the large latent heat of water. The time-averaged thrust observed in two on-orbit operations using two thruster heads was approximately 6.0 mN, with a specific impulse of 91.0 s and a power consumption of less than 14 W. The thrust-to-power ratio increased with the temperature difference between the vaporization chamber and the adjacent communication device. This indicates that the necessary power for the vaporization of the water was effectively compensated for by the heat from the surroundings owing to the appropriate thermal design of EQUULEUS. The thrust coefficient efficiency was found to be approximately 0.86, which was higher than in the ground test (~ 0.66). This is possibly due to the extremely low back pressure on orbit that mitigates the nozzle efficiency reduction due to low Reynolds number flows.

Key Words: Micropropulsion, Water Resistojet, CubeSat, Flight Operation, Artemis I

1. Introduction

In recent years, the use of small satellites has increased dramatically due to technological advancements and reduced launch expenses.^{1,2)} As the scope of space applications continues to broaden, the need for propulsion systems that can meet the increasing demands and requirements becomes more pronounced. These requirements include a range of functionalities, such as drag compensation, orbit changes, and attitude control. PROCYON, a full-scale 50 kg class space probe, showcased the potential of small satellites for deep space exploration by employing a unified propulsion system consisting of a xenon ion thruster and a cold gas jet.^{3,4)} A private company SpaceX has successfully deployed more than 1,600 Starlink constellation satellites as of 2022,⁵⁾ each equipped Hall effect thrusters operated with krypton or argon.⁶⁾ MarCO, a pair of 6U CubeSats equipped with a cold

gas jet, paved the way for interplanetary exploration of CubeSats using propulsion systems.^{7,8)}

Although several propulsion systems for small satellites have proven effective during orbital operations,^{9,10)} the choice of propellant remains predominantly around high pressure gases.¹¹⁾ As the size of the satellite decreases, the selection of propellants becomes increasingly critical.¹²⁾ In the case of CubeSats, which are usually launched as a rideshare payload, propellants that offer low pressure storage, easy handling, and high energy density are highly desirable. Considering the expanding market for small satellites and the importance of sustainable space development, the availability and cost-effectiveness of propellants should also be taken into account.

Water stands out as one of the promising propellants for small satellites due to its liquid state under standard conditions, thus eliminating the need for high-pressure gas systems.¹¹⁾ Its non-toxic properties, widespread availability, and affordability contribute to reducing development costs and expediting the necessary time for safety evaluations by launchers. Furthermore, water has the potential advantage in terms of *in-situ* resource utilization in future missions.^{13,14)} Several water-based propulsion systems have been proposed, including water ion thrusters,^{15–17)} water Hall thrusters,^{18,19)} water radio-frequency electrothermal thrusters,^{20,21)} water-



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fuelled magnetron sputtering propulsion,²²⁾ and water electrolysis propulsion.^{23–25)}

Among them, water resistojet thrusters employ a relatively straightforward thrust generation principle, which expels high-temperature water vapor to impart thrust.²⁶⁾ Effective gas-liquid separation and precise temperature management are essential for water resistojet thrusters. However, they are still challenging issues, and there have been no reports of successful water propulsion systems operating in deep space to date. For example, UK-DMC employed a water resistojet thruster²⁷⁾ but failed to generate its thrust on orbit due to inadequate separation of the liquid and gas phases. Similarly, a water resistojet thruster installed on AeroCube OCSD-7B & -7C encountered failure in thrust generation, presumably attributed to nozzle blockage caused by ice buildup.²⁸⁾

AQUARIUS (AQUA ResIstojet propUlsion System), a water resistojet propulsion system, has been developed by The University of Tokyo.^{26,29)} AQUARIUS distinguishes itself from conventional resistojet thrusters through two key features: gas-liquid separation method and power-saving strategy. Its gas-liquid separation is accomplished using a vaporization chamber, which incorporates a distinct flow path²⁹⁾ and a flow control method that will be described in detail in Section 2. Although liquid water as a propellant is valuable in small satellites, its disadvantage lies in its high latent heat. To address this, a power-saving strategy has been implemented by optimizing the configuration design to reutilize the waste heat from high temperature devices.

AQUARIUS was installed on a 6U CubeSat EQUULEUS (EQUilibriUm Lunar-Earth point 6U Spacecraft), which serves an engineering mission to demonstrate trajectory control techniques within the Sun–Earth–Moon region by a nanospacecraft.³⁰⁾ EQUULEUS was successfully launched by NASA's Space Launch System on November 16, 2022, as one of the secondary payloads of the Orion spacecraft in the Artemis I mission.³¹⁾ It successfully completed its initial checkouts of all the bus components and executed an orbit transfer using a water-propelled propulsion system in deep space for the first time.³²⁾

In this paper, we report the initial flight performance of AQUARIUS. Furthermore, a comparative evaluation of the nozzle performance on orbit is conducted in relation to the results obtained from ground testing.

2. AQUARIUS: Concept, Design, and Operation

2.1. AQUARIUS

Figure 1(a) illustrates the system diagram of AQUARIUS. Here, this system is briefly described, as details can be seen elsewhere.²⁹⁾ The propulsion system consists of a water storage unit, propellant delivery and control components, and thruster heads. The control components include a vaporization chamber, valves, and heaters. The propellant feeding is achieved through a blow-down method.

2.1.1. Water storage unit

The water storage unit comprises an aluminum tank and a bladder. The bladder contains liquid water and was placed in-

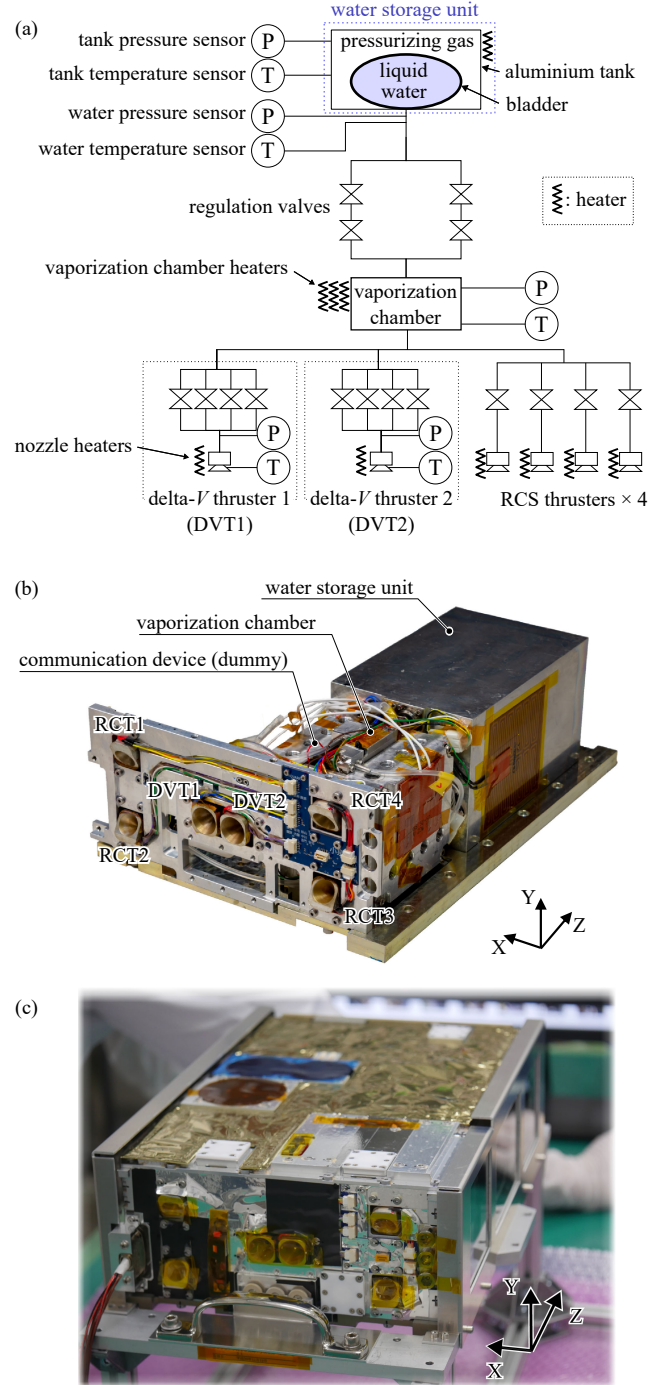


Fig. 1. (a) Schematic system diagram of AQUARIUS. (b) Photo of the AQUARIUS. Two DVTs and four RCTs are placed on the $-z$ panel of the spacecraft. The DVTs are facing toward $-z$, while the RCTs are tilted 30° with respect to the panel. (c) Photo of fully-integrated EQUULEUS, taken in May 2021, just before its shipping.

side the tank, which was initially pressurized to approximately 50 kPa by argon gas. The argon gas pressure p_{tank} was measured by the tank pressure sensor. As the pressurizing gas, any highly stable gas, such as argon or nitrogen, could be used. The pressure of liquid water stored in the bladder was also measured by the water pressure sensor, which was confirmed to be equal to p_{tank} . The temperatures of the tank T_{tank} and the liquid water T_{water} were measured using temperature sensors.

Following the water storage unit, four regulation valves are employed; two arranged in series and two in parallel. During thruster firing, two regulation valves installed in series open simultaneously, while the other two in-series regulation valves remain closed, allowing water droplets to be introduced into the vaporization chamber.

2.1.2. Vaporization chamber

The vaporization chamber serves the crucial purpose of achieving optimal gas-liquid separation. It consists of two essential components: a dedicated space for water droplet vaporization and 3D-printed labyrinthine flow paths.²⁹⁾ The injected water droplets are effectively captured by the chamber walls due to their surface tension, facilitating efficient thermal connectivity. The complex winding flow paths are designed to prevent downstream escape of water droplets to the nozzles.

The vaporization chamber temperature, T_{vap} , is maintained at approximately 31°C using three bang-bang-controlled heaters attached to the chamber. By operating at this comparatively low temperature, we could reuse the waste heat from the adjacent communication devices, compensating for the part of the latent heat of the water. The vaporization chamber pressure p_{vap} was monitored by two pressure sensors, from which we could estimate the presence of water remaining in the chamber. The subsequent water replenishment process is initiated once the onboard computer determines that no residual water is present in the chamber by p_{vap} . Therefore, the duty cycle of this pulsed operation, Δt_{cycle} , was determined from the behavior of p_{vap} .

2.1.3. Thruster heads

The water vapor produced in the vaporization chamber flows into the thruster valves. AQUARIUS employs two delta-V thrusters (DVT1 and 2) for trajectory control and four reaction control thrusters (RCT1, 2, 3, and 4) for attitude control [Fig. 1(b)]. Regarding the nozzle control temperatures, the specific impulse increases with the nozzle temperature according to the nozzle theory. However, in the development of this propulsion system, this effect was not deemed highly significant. The ground experiments have shown that the increase in specific impulse due to nozzle temperature is relatively marginal for this resistojet. In fact, in our ground tests, we varied the nozzle temperatures from 65°C to 75°C, but no discernible changes in the specific impulse were detected, while the power consumption of the nozzle heater increased by approximately 15%, corresponding to around 0.15 W per nozzle. From another perspective, the nozzle temperatures must be sufficiently higher than T_{vap} to prevent condensation of water vapor. Conversely, too high nozzle temperatures can lead to difficulty in the material selection for the components near the nozzles and damage to the satellite's structure or adhesive around the nozzle, thereby increasing the risk for the satellite's mission. Considering the trade-off among these factors, the nozzle control temperature was set at approximately 70°C. The pressures upstream of the DVT nozzles, p_{DVT1} and p_{DVT2} , were measured by pressure sensors.

It would be ideal if the center of gravity aligned precisely with the thrust vectors of the DVTs; however, it is impossible to accomplish this in the actual setup due to the misalignment

of the nozzles and the gradual shift in the center of gravity arising from propellant consumption. As a consequence, an unfavorable torque can be generated around the X-, Y- and Z-axes. To manage the torque around the Y-axis, the onboard computer automatically selected one of the DVTs to be used considering the angular momentum of the satellite during thruster firing in orbital maneuvers. The torques around the X- and Z-axes were automatically adjusted using a pair of RCTs.

2.2. Integration, initial checkouts on orbit, and orbital maneuver operations

The propulsion system was integrated into the 6U CubeSat EQUULEUS [Fig. 1(c)]. The satellite weighed approximately 10.3 kg, including a propellant liquid water mass of 1.2 kg. It underwent extensive qualification to meet the requirements imposed by in-space environment and launch vehicle specifications, as well as performance tests at ground facilities such as a vacuum chamber.^{29,33)} EQUULEUS was delivered to NASA in July 2021 and launched onboard NASA's Space Launch System (SLS) on 16 November 2022, successfully entering the lunar transfer orbit. After the delivery, satellite operation training was conducted using a suite of software and hardware simulators, spanning approximately one year. The objective was to expedite the operation timelines and refine the operation procedure. We may note that, during the design phase, we did not anticipate preservation for as long as 16 months. Potential concerns associated with prolonged storage primarily focused on the possibility of changes in p_{tank} due to gas leakage. Given the designed value of p_{tank} of 50 kPa, which is lower than atmospheric pressure, there was no need to be concerned about a decrease in p_{tank} due to outward leakage that could lead to insufficient pressure to push the liquid water. Experimental verification indicated that the inward leakage rate was negligible over several years. In fact, p_{tank} measured just before the satellite shipping was nearly identical to that measured in orbit when communications were established for the first time. Although liquid water was also present in the flow path to the regulation valves and inside the valves, there was no risk of corrosion. This was because the flow path was made of stainless steel, and the channels inside the valves were made of resin.

Initial checkouts of AQUARIUS were quickly completed within several time frames of approximately 6.5 hours in total, which is summarized in Table 1. During the checkout phase, we confirmed that all valves and heaters of AQUARIUS were functioning properly and that the torque and thrust generated were sufficient for the mission requirements of EQUULEUS. We also estimated the tilt angles of the thrust vectors for DVT1 and DVT2, which were found to be only 0.91°.

After the initial checkout, EQUULEUS executed its first orbital maneuver (delta-V operation), the DV1 operation, which aimed to raise the perilune altitude. As a result of the DV1 operation, the satellite effectively remained in the Earth-Moon system after the subsequent lunar flyby. Following the DV1 operation, the first trajectory control maneuver, the TCM1 operation, was performed to further fine-tune the

satellite's trajectory. The satellite experienced different equilibrium temperatures, with relatively high temperatures observed during the DV1 operation and lower temperatures during the TCM1 operation. The effect of the temperature conditions on AQUARIUS will be discussed in the following sections. Telemetries related to AQUARIUS acquired in the initial operation of EQUULEUS are summarized in Table 2 (including the telemetries that were not described here).

3. On-orbit Results

3.1. Time evolution of the measured parameters

Figures 2(a)(b) show the temporal evolution of p_{vap} , p_{DVT1}

Table 1. Sequence of the initial checkout, the DV1 and TCM1 operations.

Time (UTC), date	Event
06:47, 16, Nov.	Launch of the SLS
10:28, 16, Nov.	Separation from the SLS
10:32, 16, Nov.	First telemetry acquired
21:26, 16, Nov.	AQUARIUS checkout start*
21:41, 16, Nov.	Thruster valves checkout*
22:01, 16, Nov.	Heaters checkout*
22:40, 16, Nov.	Regulation valves checkout*
23:33, 16, Nov.	RCS's torque measurement*
00:53, 17, Nov.	DVT's thrust vector measurement*
03:50, 17, Nov.	AQUARIUS checkout end*
23:14, 18, Nov.	DV1 operation start
02:35, 19, Nov.	DV1 operation end
01:45, 20, Nov.	TCM1 operation start
02:10, 20, Nov.	TCM1 operation end
16:25, 21, Nov.	The first lunar closest approach

*Initial checkout.

and p_{DVT2} in the DV1 [Fig. 2(a)] and TCM1 [Fig. 2(b)] operations. The sampling rates for the pressure and temperature data were set at 1 Hz, while the sampling rate for the power data was 0.3 Hz. Immediately after the injection, water droplets began to evaporate inside the vaporization chamber,

Table 2. List of the telemetries relating to AQUARIUS acquired in the initial operation of EQUULEUS.

Parameter	Nomenclature
Tank pressure	p_{tank}
Tank temperature	T_{tank}
Water pressure	(N/A)
Water temperature	T_{water}
Vaporization chamber pressure 1	p_{vap}
Vaporization chamber pressure 2	(N/A)
Vaporization chamber temperature 1	T_{vap}
Vaporization chamber temperature 2	(N/A)
DVT1 nozzle inlet pressure	p_{DVT1}
DVT2 nozzle inlet pressure	p_{DVT2}
DVT1 nozzle inlet temperature	T_{DVT1}
DVT2 nozzle inlet temperature	T_{DVT2}
RCT1 nozzle inlet temperature	(N/A)
RCT2 nozzle inlet temperature	(N/A)
RCT3 nozzle inlet temperature	(N/A)
RCT4 nozzle inlet temperature	(N/A)
DVT1 tube* temperature	(N/A)
DVT2 tube* temperature	(N/A)
RCT1 tube* temperature	(N/A)
RCT2 tube* temperature	(N/A)
RCT3 tube* temperature	(N/A)
RCT4 tube* temperature	(N/A)
Communication device temperature	T_{comm}

*Vapor feed lines between the thruster valves and thruster heads.

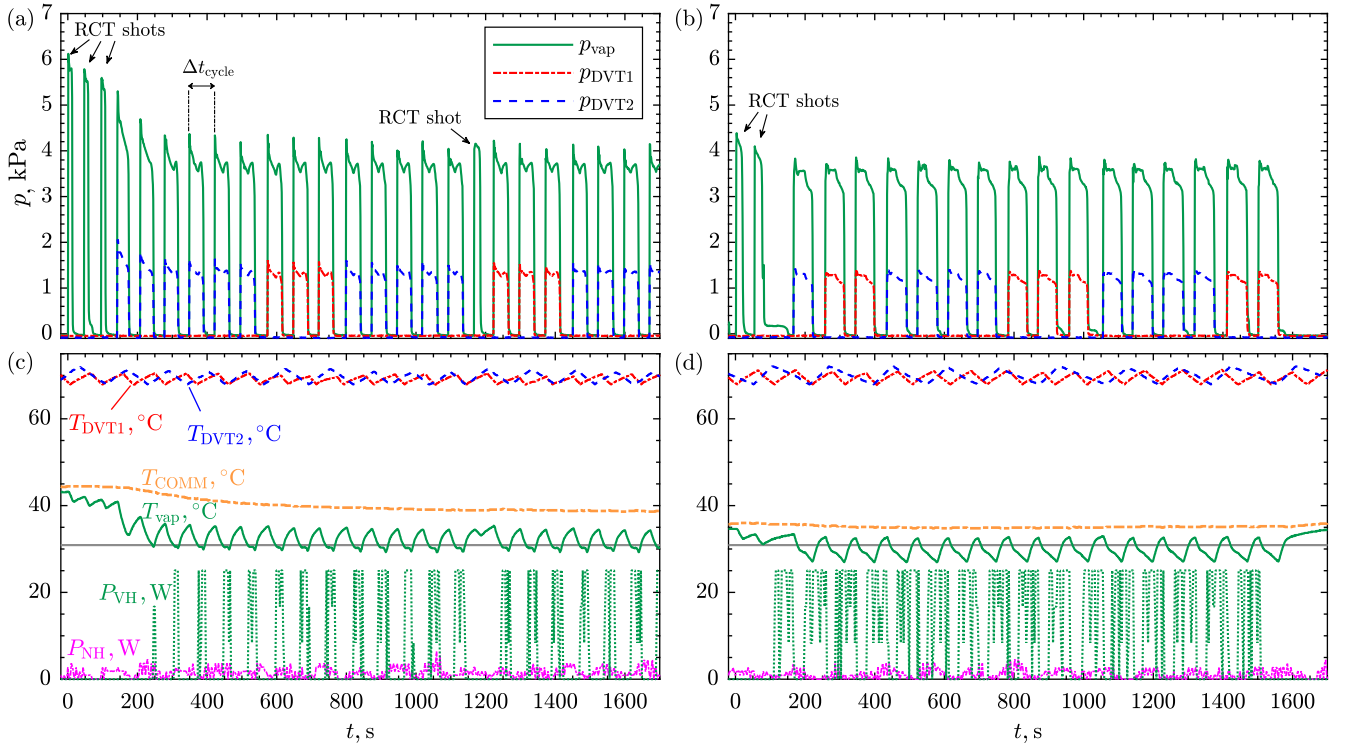


Fig. 2. Typical time history of vaporization chamber pressure p_{vap} , DVT1 nozzle pressure p_{DVT1} , and DVT2 nozzle pressure p_{DVT2} in the (a) DV1 and (b) TCM1 operations. Typical temporal history of the nozzle temperatures of the DVT1 and DVT2, T_{DVT1} and T_{DVT2} (dotted red and dashed blue lines), vaporization chamber temperature T_{vap} (solid green line), communication device temperature T_{COMM} (dashed orange line), the nozzle heater power P_{NH} (dashed magenta line), and the vaporization chamber heater power P_{VH} (dotted green line) in the (c) DV1 and (d) TCM1 operations. The gray line shows the threshold of the bang-bang control for T_{vap} .

causing the chamber pressure to rapidly increase to approximately the saturation vapor pressure at the vaporization chamber temperature (approximately 4.7 kPa at 32°C). p_{vap} , p_{DVT1} and p_{DVT2} rapidly decrease to zero as most of the water vapor is exhausted from the chamber. The onboard computer continuously monitored p_{vap} as previously described and initiated the next injection when it determined that no water droplets remained in the vaporization chamber. The total number of shots, including the DVTs and RCTs, amounted to 154 in the DV1 operation and 17 in the TCM1 operation. In the DV1 operation, the initial three shots were the RCT firings aimed at adjusting the satellite's angular momentum, while in the TCM1 operation, two RCT shots were employed for the same purpose. Following those initial RCT shots, the DVT shots started, with rapid increases of p_{DVT1} or p_{DVT2} . The cycle time Δt_{cycle} was 75.7 ± 2.9 s in the DV1 operation and 89.1 ± 3.1 s in the TCM1 operation, respectively. Δt_{cycle} was optimized to maximize the time-averaged thrust, larger than the required value from the mission of 4 mN, under the constraint of the time-averaged electric power provided by the spacecraft. The cycle is divided into two phases: the water-evaporation phase and the temperature-recovering phase. During the water evaporation phase, T_{vap} decreases due to the latent heat of the water, as seen in Figs. 2(a)(b). The temperature-recovering phase until the next injection plays a crucial role in determining T_{vap} at the beginning of the subsequent cycle. If this phase is too long, the vaporization chamber temperature recovers sufficiently (up to the bang-bang control threshold of 31°C), but this results in a reduction in the time-averaged thrust. Conversely, if the phase is too short, repeated operation lowers the time-averaged T_{vap} during the water-evaporation phase, reducing the water vapor flow rate and consequently leading to a lower time-averaged thrust.

Figures 2(c)(d) show the temporal history of the temperatures of the DVT nozzles (T_{DVT1} and T_{DVT2}), the communication devices (T_{COMM}), and the wall of the vaporization chamber (T_{vap}). The figures also display the input power to the vaporization chamber heater P_{VH} and the nozzle heater P_{NH} in the DV1 [Fig. 2(c)] and TCM1 [Fig. 2(d)] operations. The temperatures, pressures, and power almost reached a steady state after approximately the sixth and fourth shots in the DV1 and TCM1 operations, respectively. This figure clearly shows that the temperature management of AQUARIUS on orbit was successful, where the temperatures of the DVT nozzles and the vaporization chamber remained around their control values, which were approximately 70°C and approximately 31°C, respectively. Notably, due to differences in satellite attitude and resulting thermal conditions during the two operations, the average value of T_{vap} during the DV1 operation was slightly higher than in the TCM1 operation.

3.2. Measurement of ΔV and thrust coefficient estimation

All thruster operations were conducted in visible time in the DV1 and TCM1 operations. Owing to this, we could measure the velocity increment ΔV of EQUULEUS using a two-way Doppler shift in radio wave frequency, which cor-

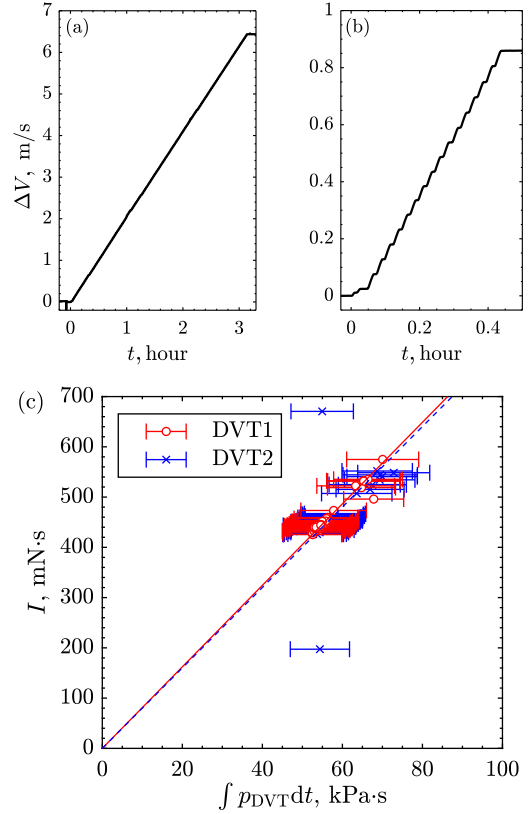


Fig. 3. Measured ΔV by the Doppler shift in the (a) DV1 and (b) TCM1 operations. (c) Correlation between the temporal integral of the nozzle pressure p_{DVT} in each shot and the impulse I measured in the DV1 and TCM1 operation. The circles and crosses show those of the DVT1 and DVT2, respectively. The solid red and dotted blue lines are the fitted lines for the DVT1 and DVT2 data, where the slopes of the line give $C_{F,\text{meas}}$.

responds to the line-of-sight speed of the spacecraft from Earth. Figures 3(a)(b) show the measured ΔV during the DV1 and TCM1 operations. The total ΔV acquired in the DV1 and TCM1 operations were approximately 6.4 m/s and 84 cm/s, respectively. The differences between the observed Doppler data and the calculated values (so-called O-C) provided the shot-by-shot impulses imparted by AQUARIUS, I_n , where n is the shot number in each operation. We may note that, to calculate I_n , we ignored the change in satellite mass because each mass shot (approximately 0.5 g; see Section 3.4) was sufficiently smaller compared to the satellite mass (approximately 10 kg). According to the nozzle theory, a thrust is expressed as $F = p_{\text{DVT}} A_t C_F$, where A_t and C_F are the throat area and the thrust coefficient, respectively. Here, p_{DVT} is p_{DVT1} or p_{DVT2} . Assuming that C_F does not depend on time t , we can obtain the measured thrust coefficient $C_{F,\text{meas}}$ using the measured I_n and p_{DVT} as

$$I_n = A_t C_{F,\text{meas}} \int_{\tau_n}^{\tau_{n+1}} p_{\text{DVT}}(t) dt, \quad (1)$$

where τ_n is the time when the n -th shot begins. Taking into account the accuracy of the pressure measurement (± 0.1 kPa), $C_{F,\text{meas}}$ were deduced to be 1.53 ± 0.03 for the DVT1 and 1.51 ± 0.02 for the DVT2, respectively [Fig. 3(c)], showing no clear difference in the performance of the two nozzles.

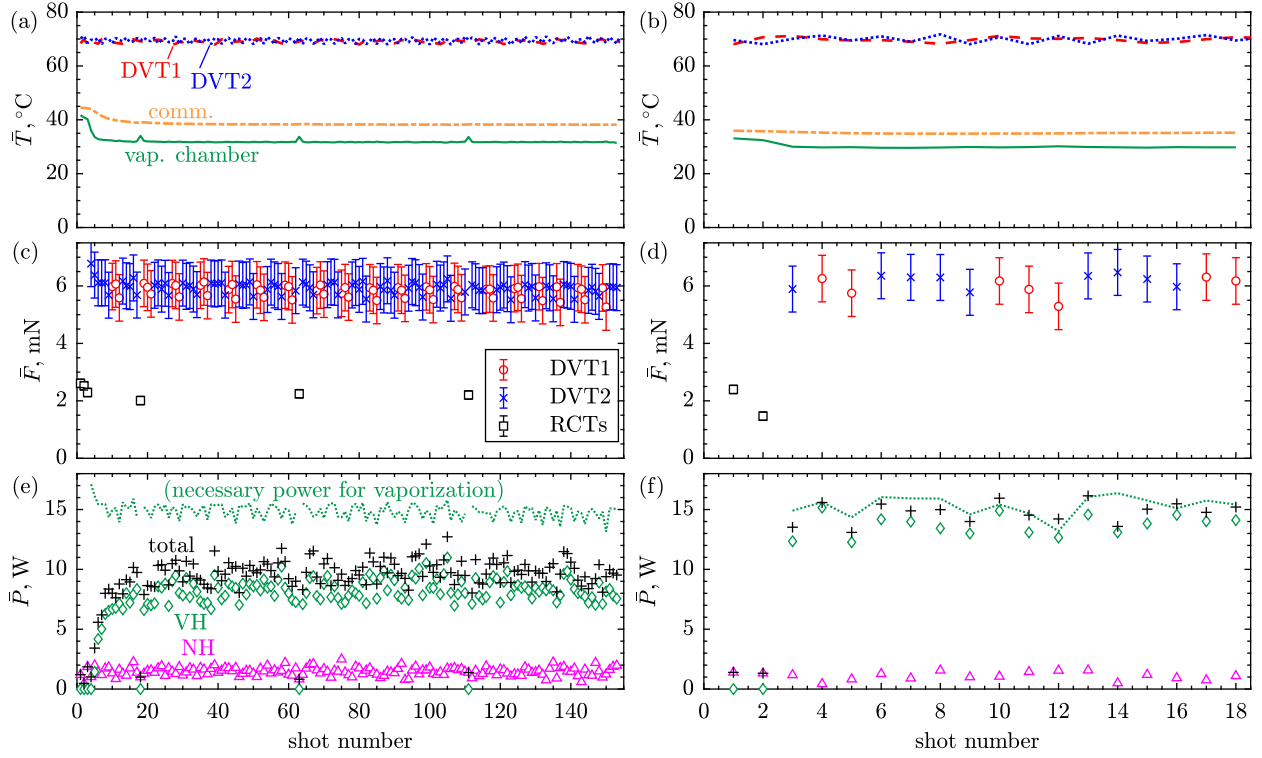


Fig. 4. Time-averaged temperatures of DVT1 (dashed red line) and DVT2 (dotted blue line) nozzles, the communication devices (dashed orange line), and the vaporization chamber (solid green line) as a function of the shot number in the (a) DV1 and (b) TCM1 operations. Measured thrusts using the DVT1, DVT2, and RCTs in the (c) DV1 and (d) TCM1 operations. Time-averaged power input to the vaporization chamber heaters (green diamonds), the nozzle heaters (magenta triangles), and the propulsion system (black crosses) in the (e) DV1 and (f) TCM1 operations. The dotted green line shows the calculated necessary power for vaporizing the injected water into the vaporization chamber, where the difference between this line and the diamonds indicates the energy supplied from the surroundings.

3.3. Thrust performance

In the following, the term “cycle-averaged” denotes the temporal average over Δt_{cycle} . Figures 4(a)(b) show the cycle-averaged temperatures of the DVT1 nozzle ($\bar{T}_{\text{DVT1}}$), DVT2 nozzle ($\bar{T}_{\text{DVT2}}$), communication devices ($\bar{T}_{\text{comm}}$), and vaporization chamber ($\bar{T}_{\text{vap}}$) as a function of the shot number. As shown here, $\bar{T}_{\text{DVT1}}$ and $\bar{T}_{\text{DVT2}}$ were successfully controlled at approximately 70°C regardless of the attitude of the satellite. $\bar{T}_{\text{comm}}$ and $\bar{T}_{\text{vap}}$ exhibited some sensitivity to the attitude of the satellite, where they were slightly higher in the DV1 operation than in the TCM1 operation.

Figures 4(c)(d) show the cycle-averaged thrust $\bar{F}$ imparted by the DVT1, DVT2, and RCTs as a function of the shot number. No detectable changes in $\bar{F}$ were observed between the two different temperature conditions, where they were 6.0 ± 0.2 mN. With these thrust values, the successful orbital maneuver of EQUULEUS was accomplished.³¹⁾

Figures 4(e)(f) show the cycle-averaged power consumed by the nozzle heaters $\bar{P}_{\text{NH}}$ (“NH”), vaporization heaters $\bar{P}_{\text{VH}}$ (“VH”), and the cycle-averaged total input power to the propulsion system $\bar{P}$ (“total”) as a function of the shot number in the DV1 and TCM1 operations. Here, the “total” power, $\bar{P}$, is defined as $\bar{P} = \bar{P}_{\text{VH}} + \bar{P}_{\text{NH}}$ and does not include the valve power. These figures clearly show that $\bar{P}_{\text{VH}}$ accounts for most of the power consumption in this propulsion system, highlighting the large latent heat of liquid water. The differences in the values of $\bar{P}_{\text{VH}}$ in the two operations arose from

the attitude of the satellite as we performed the temperature control.

During the first several shots, $\bar{T}_{\text{comm}}$ and $\bar{T}_{\text{vap}}$ were not in steady state and decreased significantly from shot to shot [Fig. 4(c)(d)], because most of the water’s latent heat is supplied by the thermal capacity of the vaporization chamber. Consequently, $\bar{P}_{\text{VH}}$ was substantially lower during this period. In a steady state, $\bar{P}$ in the DV1 and TCM1 operations were ~ 9.2 and ~ 14.8 W, respectively. Here, $H_{\text{vap}} \Delta m / \Delta t_{\text{cycle}}$ gives the cycle-averaged necessary power for vaporization of the water injected into the vaporization chamber (indicated by the dotted lines in Fig. 4(e)(f)), where H_{vap} is the enthalpy for vaporization of water and $H_{\text{vap}} = 2.25$ kJ/g. While $H_{\text{vap}} \Delta m / \Delta t_{\text{cycle}}$ is ~ 15 W, the average values for the power consumption of the vaporization heater were approximately 7.9 and 13.7 W, respectively. It clearly shows that the rest of the necessary power for vaporization was provided by the heat input from the surrounding environment, as aimed at the arrangement design of AQUARIUS.

The heat input to the vaporization chamber should depend on the temperature difference between the chamber and the surroundings. Figure 5 shows the thrust-to-power ratio $\bar{F} / \bar{P}$ as a function of $\bar{T}_{\text{comm}} - \bar{T}_{\text{vap}}$ in the two operations. This figure clearly shows that $\bar{F} / \bar{P}$ increased with the surrounding temperature, indicating that part of the latent heat of the water was effectively compensated for by the heat from the surroundings, aligning with the design concept. It is also no-

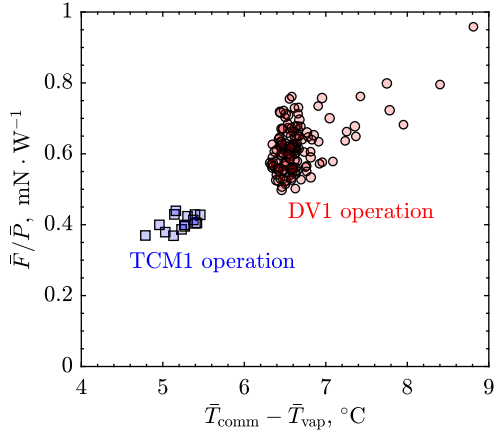


Fig. 5. Measured thrust-to-power ratio $\bar{F}/\bar{P}$ as a function of the temperature difference $\bar{T}_{\text{comm}} - \bar{T}_{\text{vap}}$. Circles and squares show the data obtained in the DV1 and TCM1 operations, respectively. The error bars are omitted here for clarity, where the typical values are approximately $\pm 0.05 \text{ mN/W}$.

table that only approximately 2°C changes in $\bar{T}_{\text{comm}} - \bar{T}_{\text{vap}}$ made a significant ($\sim 50\%$) difference in $\bar{F}/\bar{P}$, highlighting the importance of thermal design in this resistojet thruster.

3.4. Mass shot estimation and specific impulse

An accurate estimation of the injected water mass is crucial for evaluating the specific impulse of AQUARIUS. However, it is challenging because the satellite's mass is generally unknown on orbit. To ensure reliable estimates of the mass shot, we employed the following three methods.

Method (i) The tank pressure p_{tank} is the pressure of the argon gas sealed in the tank and can be considered to change depending on the water volume in the bladder according to the equation of state. Therefore, assuming that the temperature of the argon gas is equal to T_{tank} , the volume of the argon gas at the end of the n -th shot $V_{\text{argon},n}$ can be expressed as

$$V_{\text{argon},n} = \frac{T_{\text{tank}}/p_{\text{tank}}}{T_{\text{tank},0}/p_{\text{tank},0}} V_{\text{argon},0}, \quad (2)$$

where $T_{\text{tank},0}$, $p_{\text{tank},0}$ and $V_{\text{argon},0}$ are the initial temperature, pressure, and volume of the argon gas in the tank, respectively, which were measured before the start of each operation. Then the total water mass used at the end of the n -th shot, m_n , can be calculated as

$$m_n = \rho_0(V_{\text{argon},n} - V_{\text{argon},0}), \quad (3)$$

where ρ_0 is the density of liquid water (0.997 g/cm^3).

Method (ii) The integral of the vaporization chamber pressure over time is related to the impulse generated by the produced water vapor and, thus, to the mass of the injected water. Therefore, the mass shot Δm_n can be estimated as

$$\Delta m_n = \alpha \int_{\tau_n}^{\tau_{n+1}} p_{\text{vap}}(t) dt, \quad (4)$$

where α is a constant. Before the satellite launch, we tuned the value α in the ground experiment by comparing Δm_n calculated using Eq. (4) with the change in sat-

ellite mass measured using an electronic balance.

Method (iii) As the argon gas pushes the liquid water into the vacuum vaporization chamber, the mass shot should be related to p_{tank} and the conductance of the flow path from the bladder outlet to the vaporization chamber inlet. Here, the conductance and viscosity of the water depend on the temperature. Therefore, we used an empirical expression as³³⁾

$$\Delta m_n = \rho_0 a p_{\text{tank}}^b(\tau_n) T_{\text{water}}^c(\tau_n) \Delta t_{\text{RV}}, \quad (5)$$

where a , b , and c are the fitting parameters, and Δt_{RV} is the open duration time of the regulation valves. Similarly to Method (ii), the three fitting parameters were tuned using an electronic balance in the ground test. In establishing Eq. (5), our foremost objective was to accurately predict Δm_n for the EQUULEUS mission rather than strictly adhering to physical principles. In fact, the expression demonstrated the Δm_n prediction with a few percent accuracy in ground tests. The value of b was found to be 0.70, relatively close to 0.5, which is predicted by the incompressible fluid theory. The fitting parameter c was introduced to account for the temperature dependence of the water viscosity and the conductance of the flow path. The value c was optimized to 0.24, providing the best fit to the measured mass shot data obtained in the ground test. In the flight operation, Δt_{RV} was set to the value that gave the nominal mass shots of 0.63 g for the DVTs and 0.17 g for the RCTs in the ground experiment.

Figures 6(a)(b) show p_{tank} and T_{tank} with respect to the shot number in the DV1 and TCM1 operations. It should be noted that, due to the limited pressure resolution of the tank pressure sensor (0.3 kPa), the changes in p_{tank} for each shot could not be precisely detected. As a result, step decreases in p_{tank} were observed when the changes in p_{tank} exceeded 0.3 kPa.

Figures 6(c)(d) show the total water mass used until the n -th shot, m_n , estimated using Methods (i), (ii), and (iii) in the DV1 and TCM1 operations. Here, m_n is defined as

$$m_n = \sum_{k=1}^n \Delta m_k, \quad (6)$$

where k is the natural number. As is seen in Fig. 6(c), m_n estimated by Method (i) and Method (ii) exhibit very similar values, whereas the estimated values using Method (iii) deviate from them by approximately 20% in the DV1 operation. On the other hand, in the TCM1 operation, m_n estimated by all three methods shows very good agreement.

The estimation of the mass shot can be validated by comparing it with the theoretical value Δm_{th} . Figures 6(e)(f) show the cycle-averaged discharge coefficient $\bar{C}_d$, which is defined as

$$\bar{C}_d \equiv \frac{\Delta m}{\Delta m_{\text{th}}}, \quad (7)$$

where Δm_{th} is

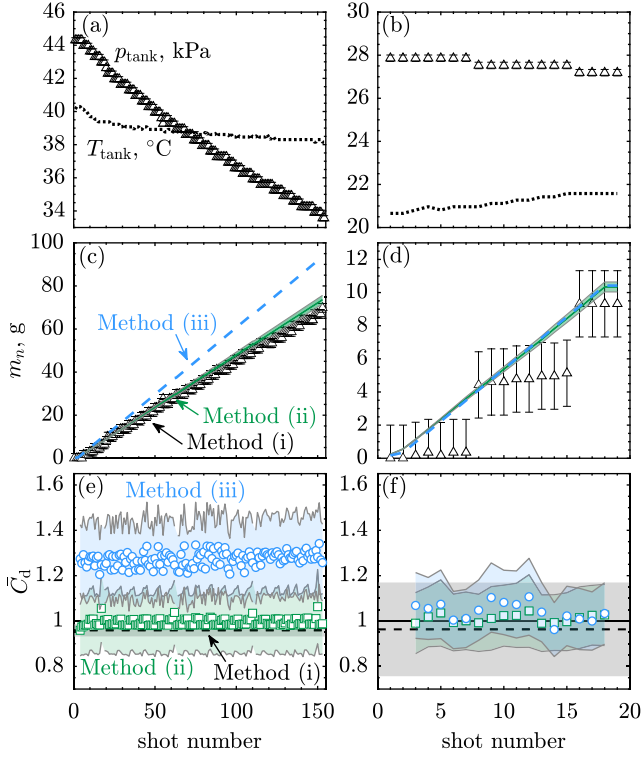


Fig. 6. Tank pressure p_{tank} (triangle) and tank temperature T_{tank} (dotted line) as functions of the shot number in the (a) DV1 and (b) TCM1 operations. Estimated total propellant mass used m_n as a function of the shot number in the (c) DV1 and (d) TCM1 operations. The triangles, solid green lines, and dashed cyan lines show the estimated values using Methods (i), (ii), and (iii), respectively. The shaded green region indicates the uncertainty in Methods (iii), where the uncertainty in Methods (iii) is omitted here due to the small value (typically 0.8%). Discharge coefficient $\bar{C}_d$ as a function of the shot number in the (e) DV1 and (f) TCM1 operations. The dashed black lines, squares, and circles show $\bar{C}_d$ estimated with Methods (i), (ii), and (iii), respectively. The shaded region indicates the uncertainties for each value.

$$\Delta m_{\text{th}} = \frac{\bar{F} \Delta t_{\text{cycle}}}{C_{\text{th}}^* C_{F,\text{th}}}. \quad (8)$$

C_{th}^* is the theoretical characteristic exhaust velocity defined as

$$C_{\text{th}}^* = \sqrt{\frac{R \bar{T}_{\text{DVT}}}{\gamma} \left(\frac{\gamma+1}{2} \right)^{(\gamma+1)/(\gamma-1)}}, \quad (9)$$

where R and γ are the gas constant and specific heat ratio of water, respectively, and $\bar{T}_{\text{DVT}}$ indicates $\bar{T}_{\text{DVT1}}$ or $\bar{T}_{\text{DVT2}}$. Here, we assumed that the stagnant water vapor temperature at the nozzle inlet was equal to the nozzle temperature. $C_{F,\text{th}}$ is the theoretical thrust coefficient. Assuming a zero back pressure, an isentropic expansion, an ideal gas, and a non-viscous quasi-one-dimensional flow, it can be written as³⁴⁾

$$C_{F,\text{th}} = \sqrt{\frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \left[1 - \left(\frac{p_{\text{exit}}}{p_{\text{DVT}}} \right)^{\frac{\gamma-1}{\gamma}} \right]} + \frac{p_{\text{exit}}}{p_{\text{DVT}}} \frac{A_{\text{exit}}}{A_t}, \quad (10)$$

where

$$\frac{A_{\text{exit}}}{A_t} = \sqrt{\left(\frac{\gamma-1}{2} \right) \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \left[\left(\frac{p_{\text{exit}}}{p_{\text{DVT}}} \right)^{\frac{2}{\gamma}} - \left(\frac{p_{\text{exit}}}{p_{\text{DVT}}} \right)^{\frac{\gamma+1}{\gamma}} \right]^{-1}}, \quad (11)$$

and A_{exit} and p_{exit} are the nozzle exit area, the nozzle exit pressure, respectively, and p_{DVT} indicates p_{DVT1} or p_{DVT2} . It should be noted that it is difficult to calculate the shot-by-shot $\bar{C}_d$ due to the low resolution of the analog-to-digital conversion in p_{tank} (0.3 kPa) used in Method (i). Therefore, as reference values, $\bar{C}_d$ is calculated in Method (i) using only the two values of p_{tank} : those measured just before the operation and after all shots in the operation. $\bar{C}_d$ is approximately unity when Methods (i) and (ii) are used in both the DV1 and TCM1 operations. On the other hand, $\bar{C}_d$ exceeding unity is obtained using Method (iii) in the DV1 operation, while it is approximately unity in the TCM1 operation. We consider Method (i) to be the most reliable due to the simple physical principle and the reasonable agreement of $\bar{C}_d$ to 1. Method (ii) also appeared to be reliable, as it demonstrates a similar $\bar{C}_d$ as Method (i). Regarding Method (iii), the cause of $\bar{C}_d > 1$ in the DV1 operation is currently unclear and requires further investigation.

The cycle-averaged specific impulse $\bar{I}_{\text{sp}}$ is defined by

$$\bar{I}_{\text{sp}} \equiv \frac{\bar{F} \Delta t_{\text{cycle}}}{g_0 \Delta m}, \quad (12)$$

where g_0 is the gravitational acceleration. Using Δm estimated by Methods (i) and (ii), $\bar{I}_{\text{sp}}$ were estimated to be 91.1 ± 1.7 (DVT1) and 92.0 ± 1.3 s (DVT2) in the DV1 operation and 89.4 ± 4.6 (DVT1) and 91.0 ± 3.9 s (DVT2) in the TCM1 operation, respectively. The specific impulse efficiency, $\eta_{I_{\text{sp}}}$, defined as the ratio of $\bar{I}_{\text{sp}}$ to the theoretical specific impulse $I_{\text{sp,th}}$, defined as $I_{\text{sp,th}} = C_{\text{th}}^* C_{F,\text{th}} / g_0$, was 0.86 ± 0.01 . Since the specific impulse can be drastically reduced due to the ejection of water droplets from the nozzle, this relatively high $\eta_{I_{\text{sp}}}$ implies no critical failure in gas-liquid separation, which has been one of the significant challenges in conventional water resistojet thrusters. This result also shows that $\eta_{I_{\text{sp}}}$ improved from the ground test, where it was approximately 0.66.²⁹⁾ The performance of AQUARIUS is summarized in Table 3, together with the data obtained in the ground test.³⁵⁾

3.5. Nozzle performance analysis

Here, we discuss the detailed nozzle performance. In this section, we use the cycle-averaged values for all parameters without overlines. For simplicity, we focus on the DVT1 data, as there was no critical difference between the performance of the DVT1 and DVT2 nozzles.

Figure 7 shows $C_{F,\text{th}}$ as a function of $p_{\text{exit}}/p_{\text{DVT1}}$. The ideal thrust coefficient $C_{F,\text{id}}$ was determined using the geometric aperture ratio of 47.4²⁹⁾ as the value of A_{exit}/A_t . The thrust coefficient measured using a thrust stand in the ground test was ~ 1.2 , which did not yield a solution for $p_{\text{exit}}/p_{\text{DVT1}}$ as it was smaller than the theoretical minimum. It indicates that the background pressure had significantly degraded the thrust coefficient of this low-Reynolds-number

Table 3. DVT performance summary of AQUARIUS. The uncertainties correspond to the standard deviations.

Parameter	Ground test	DV1 (high- T operation)	TCM1 (low- T operation)
Cycle time, Δt_{cycle}	98.4 ± 4.5 s	75.7 ± 2.9 s	89.1 ± 3.1 s
Vaporization chamber pressure, $\bar{p}_{\text{vap}}$	2.0 ± 0.1 kPa	2.0 ± 0.1 kPa	2.0 ± 0.3 kPa
Vaporization chamber temperature, $\bar{T}_{\text{vap}}$	$28.2 \pm 0.9^\circ\text{C}$	$32.0 \pm 1.4^\circ\text{C}$	$30.2 \pm 1.0^\circ\text{C}$
Vaporization heater power, $\bar{P}_{\text{VH}}$	13.7 ± 9.4 W	7.9 ± 1.6 W	13.7 ± 0.5 W
Mass shot, Δm	0.60 ± 0.02 g	0.50 ± 0.04 g	0.61 ± 0.04 g
Mass flow rate, $\dot{m}$	6.1 ± 0.2 mg/s	6.5 ± 0.2 mg/s	6.8 ± 0.4 mg/s
Thrust, $\bar{F}$	4.1 ± 0.4 mN	5.9 ± 0.2 mN	6.1 ± 0.2 mN
Specific impulse, $\bar{I}_{\text{sp}}$	68.5 ± 5.5 s	91.7 ± 1.0 s	90.3 ± 3.0 s

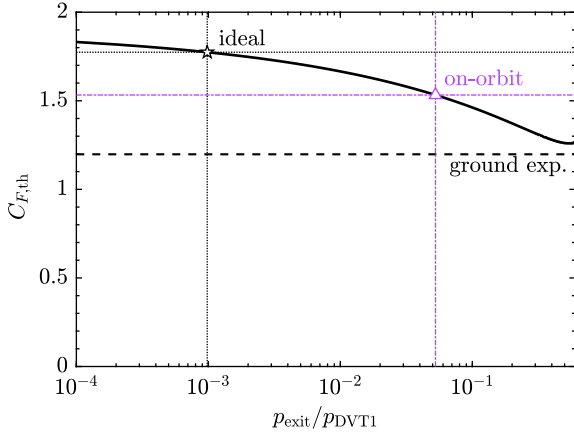


Fig. 7. Theoretical thrust coefficient $C_{F,\text{th}}$ as a function of the pressure ratio at the nozzle exit and the nozzle inlet $p_{\text{exit}}/p_{\text{DVT1}}$ (solid black). The thrust coefficient at the “ideal” (black pentagram) point, $C_{F,\text{id}}$, was deduced from $p_{\text{exit}}/p_{\text{DVT1}}$ calculated by the geometric nozzle aperture ratio. On the other hand, for the “on-orbit” case (purple triangle), $p_{\text{exit}}/p_{\text{DVT1}}$ was deduced using $C_{F,\text{meas}}$ by solving Eq. (10). The dashed line indicates the measured thrust coefficient obtained in the ground test.

nozzle in the ground experiment, as suggested in a number of previous studies.^{34–37)} It has also been suggested that the degradation of the thrust coefficient is mitigated under very low back pressure conditions.³⁵⁾ During ground testing, the chamber pressure during thruster operation was typically on the order of 10^{-2} Pa. On the other hand, on orbit, the back pressure can be estimated to be below 10^{-8} Pa, considering that the satellite’s altitude was approximately 200,000 km during the DV1 and TCM1 operations. Given the measured thrust coefficient $C_{F,\text{meas}}$, $p_{\text{exit}}/p_{\text{DVT1}}$ on orbit was successfully estimated by solving Eq. (10) implicitly. We may note that the uncertainty in $C_{F,\text{meas}}$ was approximately 1.8%. Regarding the uncertainty in $C_{F,\text{id}}$, as A_{exit} and A_t are 17.9 and 2.6 mm, respectively, the uncertainty of A_{exit}/A_t is $\pm 5.5\%$, considering the ordinary manufacturing tolerance of ± 0.1 mm. Therefore, the uncertainty of $p_{\text{exit}}/p_{\text{DVT1}}$ is deduced to be $+12\%$ and -7.2% from Eq. (11). However, the uncertainty of $C_{F,\text{id}}$ is calculated as $\pm 0.014\%$ due to the very weak dependence of the thrust coefficient on $p_{\text{exit}}/p_{\text{DVT1}}$ [Fig. 7], which is negligibly small.

Figure 8 shows the temperature-pressure (T - p) diagram of the water vapor. The point labeled “vap. chamber” corresponds to the water vapor in the vaporization chamber. It flows into the nozzle inlet through the piping and valve, experiencing a pressure drop and being heated by the nozzle wall. When the water vapor expands isentropically from

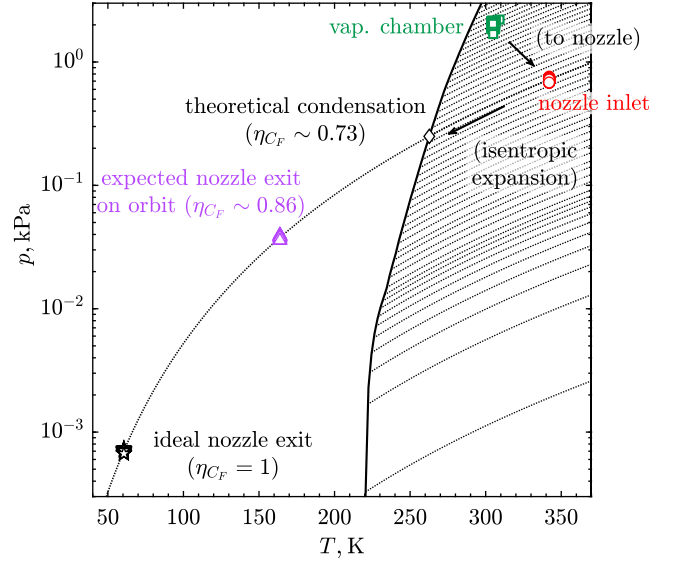


Fig. 8. T - p diagram of the water vapor in the vaporization chamber (squares) and nozzle inlet (circles) of the DVT1. The triangles indicate the estimated nozzle exit condition of the DVT1, where the pentagrams indicate the ideal nozzle exit condition. The solid line is added to show the saturated vapor pressure curve of water. The dotted lines indicate isentropic changes, where the line corresponding to Eq. (13) extends beyond the intersection with the saturated vapor pressure curve. We may note that the temperatures and pressures of the vaporization chamber and nozzle inlet are the measured values, while those of the nozzle exit are the estimated ones.

the “nozzle inlet” point, its temperature T and pressure p decrease according to Poisson’s law:

$$\frac{p^{\frac{\gamma-1}{\gamma}}}{T} = \frac{p_{\text{DVT1}}^{\frac{\gamma-1}{\gamma}}}{T_{\text{DVT1}}} \quad (13)$$

When the thrust coefficient achieves its theoretical maximum with $\eta_{C_F} = 1$, where η_{C_F} is the thrust coefficient efficiency, p_{exit} reaches its theoretical minimum. To discuss an expansion of water vapor, it is important to consider the possibility of condensation, as it can cause deviations in η_{C_F} from the ordinary nozzle theory. If all the gas condenses into a liquid within the nozzle, the gas expansion ends and the propellant is no longer accelerated. Theoretically, the isentropically expanding water vapor gas condenses when the temperature and pressure reach the intersection of the curves of the isentropic expansion and the saturated vapor pressure (labeled as “theoretical condensation” in Fig. 8). At this point, η_{C_F} is calculated to be 0.73. Since the measured η_{C_F} on orbit is large-

er than this value (~ 0.86), the estimated p_{exit} falls between the “ideal” and “condensation” points. It suggests that not all of the water vapor condensed in the nozzle on orbit. The water vapor may exist in a supercooling state, which is difficult to predict using classical thermodynamics and, thus, ordinary nozzle theory. On the other hand, the measured η_{CF} is still lower than unity. This may imply either partial condensation or the presence of non-negligible low-Reynolds-number effects.

By assuming $C_d = 1$, we can discuss the C^* efficiency η_{C^*} . The measured C^* can be calculated as $p_{DVT} A_t / \dot{m}$, and η_{C^*} is found to be approximately unity both on orbit and in the ground test. As η_{C^*} represents the discrepancy between the ideal and actual state of the gas, the water vapor temperature in the nozzle and the nozzle wall temperature can be considered almost the same in this resistojet thruster, both on orbit and in the ground test.

4. Conclusion

We analyzed the on-orbit performance results of AQUARIUS, a water resistojet propulsion system, with which a 6U CubeSat EQUULEUS successfully executed an orbit transfer using a water-propelled propulsion in deep space for the first time. The thrust and specific impulse on orbit were approximately 6.0 mN and 91.0 s, and the power consumption was less than 14 W. This indicates that the necessary power for the vaporization of the water was effectively compensated for by the heat from the surroundings owing to the appropriate thermal design of EQUULEUS. The measured thrust coefficient efficiency was approximately 0.86, which was higher than in the ground test (~ 0.66). The analysis of C_F indicates that the water vapor temperature and pressure at the nozzle exit were probably lower than those under conditions where the water vapor is condense. Extremely low back pressure on orbit is considered to mitigate the nozzle efficiency reduction due to low Reynolds number flows.

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